

OS - Permissions Programs

Tasks-

1) Give Read and Write only to the owner

```
ananya@DESKTOP-42I5F93:~$ chmod 600 sample.txt
ananya@DESKTOP-42I5F93:~$ ls -l sample.txt
-rw----- 1 ananya ananya 0 Aug 17 05:35 sample.txt
```

Owner = 6 → read (4) + write (2)

Group = 0 → nothing (no permission)

Others = 0 → nothing (no permission)

2) Give Read and Write to the owner and group

```
ananya@DESKTOP-42I5F93:~$ chmod 660 sample.txt
ananya@DESKTOP-42I5F93:~$ ls -l sample.txt
-rw-rw---- 1 ananya ananya 0 Aug 17 05:35 sample.txt
```

Owner = 6 → read (4) + write (2)

Group = 6 → read (4) + write (2)

Others = 0 → nothing (no permission)

3) Only owner can read and write

```
ananya@DESKTOP-42I5F93:~$ chmod 600 sample.txt
ananya@DESKTOP-42I5F93:~$ ls -l sample.txt
-rw----- 1 ananya ananya 0 Aug 17 05:35 sample.txt
```

Owner = 6 → read (4) + write (2)

Group = 0 → nothing (no permission)

Others = 0 → nothing (no permission)

4) Owner read/write, everyone else read-only

```
ananya@DESKTOP-42I5F93:~$ chmod 644 sample.txt
ananya@DESKTOP-42I5F93:~$ ls -l sample.txt
-rw-r--r-- 1 ananya ananya 0 Aug 17 05:35 sample.txt
```

Owner = 6 → read (4) + write (2)

Group = 4 → read (4)

Others = 4 → read (4)

5) Owner can do everything, others can only read and execute

```
ananya@DESKTOP-42I5F93:~$ chmod 755 sample.txt
ananya@DESKTOP-42I5F93:~$ ls -l sample.txt
-rwxr-xr-x 1 ananya ananya 0 Aug 17 05:35 sample.txt
```

Owner = 7 → read (4) + write (2) + execute (1)

Group = 5 → read (4) + execute (1)

Others = 5 → read (4) + execute (1)

6) Owner and group can read/write

```
ananya@DESKTOP-42I5F93:~$ chmod 660 sample.txt
ananya@DESKTOP-42I5F93:~$ ls -l sample.txt
-rw-rw---- 1 ananya ananya 0 Aug 17 05:35 sample.txt
```

Owner = 6 → read (4) + write (2)

Group = 6 → read (4) + write (2)

Others = 0 → nothing (no permission)

7) Only owner can execute

```
ananya@DESKTOP-42I5F93:~$ chmod 700 sample.txt
ananya@DESKTOP-42I5F93:~$ ls -l sample.txt
-rwx----- 1 ananya ananya 0 Aug 17 05:35 sample.txt
```

Owner = 7 → read (4) + write (2) + execute (1)

Group = 0 → nothing (no permission)

Others = 0 → nothing (no permission)

Fill in the permission numbers for the following:

8) Owner: read + write; Group: nothing; Others: nothing → **600**

9) Owner: read + write; Group: read; Others: read → **644**

10) Owner: read + write + execute; Group: read + execute; Others: read + execute → **755**

11) Owner: read + write; Group: read + write; Others: nothing → **660**

12) Everyone: read only → **444**